Harin Lee

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harinboy.github.io

Academic Interests

My research interests lie in the broad area of decision-making theory. In particular, my work focuses on designing practical algorithms with theoretically guaranteed performance in linear bandits and reinforcement learning.

Education

University of Washington, Seattle, WA, USA — Sep 2025-Present
Advisor : Prof. Kevin Jamieson
Ph.D. Program
Paul G. Allen School of Computer Science and Engineering

Seoul National University, Seoul, Korea — Mar 2018–Feb 2025
Bachelor of Science
Double Major in Computer Science and Engineering and Mathematical Sciences
GPA : Overall 4.18/4.3, CSE 4.2/4.3, Math 4.3/4.3
Leave of Absence for Military Service: Sep 2020–Aug 2022

Seoul Science High School, Seoul, Korea — Mar 2015–Feb 2018

Publications

[1] M. Oh and **H. Lee**, "Linear contextual bandits with quasi-optimism", in *The Fortieth Annual Conference on Neural Information Processing Systems*, 2026 (To appear).

[2] **H. Lee** and K. Jamieson, "Minimax optimal strategy for delayed observations in online reinforcement learning (**Oral**)", in *Proceedings of the 43rd International Conference on Machine Learning*, vol. 306, PMLR, 2026, pp. 63 800–63 838.

[3] **H. Lee** and M. Oh, "Unified framework of distributional regret in multi-armed bandits and reinforcement learning", in *Proceedings of Thirty Ninth Conference on Learning Theory*, vol. 336, PMLR, 2026, pp. 4520–4584.

[4] **H. Lee** and M. Oh, "Infrequent exploration in linear bandits", *Advances in Neural Information Processing Systems*, vol. 38, pp. 75 483–75 517, 2026.

[5] **H. Lee** and M. Oh, "Minimax optimal reinforcement learning with quasi-optimism", in *The Thirteenth International Conference on Learning Representations*, 2025.

[6] **H. Lee**, T. Hwang, and M. Oh, "Lasso bandit with compatibility condition on optimal arm", in *The Thirteenth International Conference on Learning Representations*, 2025.

[7] **H. Lee** and M. Oh, "Improved regret of linear ensemble sampling", *Advances in Neural Information Processing Systems*, vol. 37, pp. 92 803–92 831, 2025.

Experience

Amazon Ads, Seattle, WA, USA — July 2026–Nov 2026
Applied Scientist Intern in PubTech Team
- Built and evaluated bidder behavior estimation models.

Graduate School of Data Science, Seoul National University — April 2023–May 2025
Undergraduate Researcher
Advisor : Prof. Min-hwan Oh
- Designed practical algorithms across multiple settings in bandits and reinforcement learning and provided their efficient regret guarantees.

Thunder Research Group, Dept. of CSE, Seoul National University — Jul 2022–Dec 2022
Undergraduate Research Opportunity Program, Creative Integrated Design 1
Advisor : Prof. Jaejin Lee
- Developed low-precision and quantization methods for GPT-2

Reviewer
JMLR, TMLR, AAAI 2026, NeurIPS 2026, ICML 2026 (Gold reviewer), ICLR 2026, NeurIPS 2025 (Top reviewer)

Honors and Awards

Paper Awards

K-Data Science Conference — Nov 2024
- President of National Research Foundation of Korea Award (Top 2 among 28 finalists) [6]

Korean Artificial Intelligence Association Summer Conference — Aug 2024
- Excellence Paper Award (Top 7) [6]

Scholarships

KFAS Overseas PhD Scholarship — September 2025-Present
Korea Foundation for Advanced Studies

Kwanjeong Domestic Scholarship — Mar 2020–Feb 2024
Kwanjeong Educational Foundation

Programming Competitions

Silver Medal (7th Place) & Asia Pacific Champion The 47th ICPC World Finals — Apr 2024

4th Prize Samsung Collegiate Programming Cup — Sep 2023

Silver Prize (3rd Place) The 2022 ICPC Asia Seoul Regional Contest — Nov 2022

2nd Place The 2022 ICPC Asia Korea National First Round Programming Contest — Oct 2022

3rd Prize Samsung Collegiate Programming Cup — Sep 2022

Skills

Programming Languages: **C, C++, Python**

Machine Learning Frameworks: **PyTorch, TensorFlow**

Languages: **Korean** (native), **English** (fluent)